

DECK 06 · WEEK 3 · MORNING 2

FSA I

Framework, IS & BS

Reading financial statements like an analyst — from IFRS 15 revenue to goodwill impairment

CFA Level I · Vol 4 Learning Modules 1–3

Roadmap — FSA I — Framework, Income Statement & Balance Sheet

5 sections · ~2 hours · Built bottom-up from CFA readings

01

FSA framework & reporting standards

Six-phase framework · four regulated source layers · IFRS vs US GAAP · the audit opinion ladder

02

Income statement — revenue & expense recognition

IFRS 15 five steps · Armored Vehicles long-term contract · MegaDigital principal vs agent · Kahn inventory · CAP vs NOW

03

Non-recurring items & EPS

Discontinued ops · Angler weighted shares · Oppnox if-converted · Hihotech treasury stock

04

Balance sheet — classification & measurement

Current vs non-current · historical, amortized, fair value · IFRS 9 three buckets · Exhibit 13 ratios

05

Intangibles, goodwill & common-size analysis

Alpha Inc. development costs · Safeway impairment · Companies A/B/C · Apple vs Microsoft

Learning objectives

By the end of this deck you should be able to:

1. Walk through the six phases of a financial statement analysis and identify the four regulated source layers — filings, notes, MD&A, audit report — that feed each phase.
2. Apply the converged IFRS 15 / ASC 606 five-step revenue-recognition model to a long-term contract and explain how principal-vs-agent classification reshapes reported revenue.
3. Adjust reported earnings for non-recurring items and compute basic and diluted EPS under the if-converted and treasury-stock methods, including the antidilution check.
4. Identify which measurement basis — historical cost, amortized cost, or fair value — applies to a balance-sheet line, and sort financial assets into the three IFRS 9 buckets.
5. Distinguish identifiable intangibles from goodwill, capitalize the right development-phase costs under IFRS, and read a common-size statement of either kind.

SECTION 01

FSA framework & reporting standards

Financial statement analysis is a disciplined six-phase process that ends with a communicated recommendation, not with a ratio

Phase 1 is the one students skip; Phase 5 is the one they forget exists. Ratios without a question answer nothing, and analysis never communicated is not analysis.

Phase	Sources of information	Output
1 · Articulate purpose & context	Mandate, client need, guidelines	Questions to answer; timetable
2 · Collect data	Filings, IR calls, site visits	Usable data; questionnaires
3 · Process data	The phase-2 data, plus tools	Adjusted stmts, ratios, charts
4 · Analyze and interpret	Raw plus processed data	Results, forecasts, valuations
5 · Conclude and communicate	Results plus prior reports	The report and the call
6 · Follow-up	Actual vs expected; new facts	Revised forecasts and rating

WHY THE FRAMEWORK IS EXAMINABLE

It separates fact-gathering from interpretation, and interpretation from recommendation. Note where forecasts and valuations actually live: Phase 4, not Phase 3. Phase 3 only reshapes the data — common-size statements, ratios, charts. Confusing the two is the classic exam trap.

Source: CFA Institute, Curriculum Vol 4, Learning Module 1, Exhibit 1 "Financial Statement Analysis Framework", p. 6.

A complete analyst file draws on four regulated layers: the filings, the notes, management's commentary, and the auditor's report

Each layer answers a different question. The statements give the "what"; the notes give the "how"; MD&A gives the "why"; the audit gives the "can we trust it?"

- 1 Regulatory filings & financial statements**
Forms 10-K / 20-F / 40-F annually, 10-Q quarterly, 8-K for material events, DEF-14A proxy for compensation and governance. The four statements sit inside them: balance sheet, income statement, cash flows, changes in equity.
- 2 Notes & supplementary schedules**
Basis of preparation, accounting policies and estimates, segment data, contingencies, related parties. Sea Ltd.'s 2021 statements run to more than 60 pages of notes — this is where the real answers live.
- 3 Management commentary / MD&A**
The IASB Practice Statement frames five elements: nature of the business, objectives and strategy, resources and risks, results, critical performance measures. Unaudited almost everywhere — Germany is the exception.
- 4 Auditor's report**
Opinion on fair presentation, the basis for that opinion, and for listed issuers the Key or Critical Audit Matters. Reasonable assurance, never absolute — audits sample and statements contain estimates.

Source: CFA Institute, Curriculum Vol 4, Learning Module 1, "Regulated Sources of Information", pp. 13–26.

IFRS and US GAAP have converged on revenue and leases but still diverge on LIFO, write-down reversal, development costs and revaluation

Convergence is a direction of travel, not a destination. An analyst comparing cross-border peers has to isolate the residual differences before comparing anything.

Area	IFRS	US GAAP
Standard setter and style	IASB – principles-based	FASB – rules-based
Revenue recognition	IFRS 15 – the converged 5 steps	ASC 606 – the same 5 steps
Interest paid – cash flow	Operating OR financing	Operating only
Inventory – LIFO	Prohibited: FIFO or wtd-average	Permitted, widely used for tax
Inventory write-down reversal	Permitted if NRV recovers	Prohibited – write-down sticks
Internal development costs	Capitalise once criteria met	Expensed (software exception)
Intangibles – revaluation	Cost or revaluation model	Cost model only

PRACTICAL CONSEQUENCE

Revenue (IFRS 15 / ASC 606) and leases (IFRS 16 / ASC 842) are converged or close. The residue still matters: two companies with identical cash flows can report different EBITDA, book value and ROA.

Source: CFA Institute, Curriculum Vol 4, Learning Module 1, Exhibit 6 "Selected Major Differences between IFRS and US GAAP", p. 27; revaluation model per LM 3, p. 93.

The auditor's opinion sits on a four-step ladder from unqualified clean bill down to no opinion at all – read it before the numbers

Anything other than unqualified is a flag. An audit gives reasonable assurance, never absolute — it samples, and the statements it checks are full of estimates.

Opinion	What it means	Analyst action
Unqualified — "clean"	Fairly presented, in all material respects	Proceed as normal
Qualified	A scope limitation or a specific exception	Isolate it, then adjust
Adverse	Materially departs from standards – not fair	Halt the model
Disclaimer of opinion	Auditor could not form an opinion at all	Treat it as unaudited

US – THE SECOND OPINION (SOX)

Under Sarbanes-Oxley the auditor also opines on internal control over financial reporting, and management must accept responsibility for its effectiveness in writing. A material-weakness flag deserves the same weight as a qualified opinion on the numbers.

KEY AND CRITICAL AUDIT MATTERS

ISA 701 requires Key Audit Matters internationally; the PCAOB requires Critical Audit Matters in the US. Both map where judgement drives the numbers — revenue, impairment, tax. The curriculum is blunt that these are not necessarily what matters most to an investor.

Source: CFA Institute, Curriculum Vol 4, Learning Module 1, "Audit Reports", pp. 23–26.

SECTION 02

Income statement: revenue & expense recognition

IFRS 15 and ASC 606 collapse all revenue recognition into a single principles-based five-step model, applied to every contract with a customer

Core principle — recognise revenue to depict the transfer of promised goods or services in an amount reflecting the consideration the entity expects to be entitled to.

Contract → Obligations → Price → Allocate → Recognise

where: Steps 1–2 identify WHAT is promised. Steps 3–4 quantify HOW MUCH. Step 5 alone determines WHEN.

- 1 Identify the contract**
Enforceable, commercial substance, collection probable. IFRS: more likely than not. US GAAP: likely.
- 2 Identify performance obligations**
Promises to transfer DISTINCT goods or services — the customer can benefit from each on its own.
- 3 Determine the transaction price**
What the seller expects to receive. Variable consideration included only if it will not reverse.
- 4 Allocate to the obligations**
Split by relative standalone selling prices when the contract bundles several obligations.
- 5 Recognise revenue**
When (point in time) or as (over time) control transfers. Output or input methods measure progress.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, "Accounting Standards for Revenue Recognition", p. 40. IFRS 15 / ASC 606 core principle.

On a long-term contract IFRS 15 recognises revenue as the work progresses — and a revised cost estimate rewrites the profit you already booked

Over-time recognition applies when the asset has no alternative use to the seller and there is an enforceable right to payment for work completed — a bespoke weapons system qualifies.

ARMORED VEHICLES INC. · USD 10M WEAPONS SYSTEM · INPUT METHOD

Contract price USD 10.0m. Estimated total cost USD 7.0m → expected profit USD 3.0m. Two-year build. Year 1 costs USD 4.2m.

1. Over-time test met: no alternative use, enforceable right to payment for work done
2. Single performance obligation; transaction price USD 10.0m, all allocated to it
3. Input method — progress = costs incurred / total estimated costs
4. Year 1: $4.2 / 7.0 = 60\%$ → revenue $60\% \times 10.0 = \text{USD } 6.0\text{m}$, profit $60\% \times 3.0 = \text{USD } 1.8\text{m}$
5. Year 2 the estimate moves: cumulative cost comes in at USD 7.5m, not USD 7.0m
6. Year 2 revenue = $10.0 - 6.0 = \text{USD } 4.0\text{m}$; cost = $7.5 - 4.2 = \text{USD } 3.3\text{m}$ → profit USD 0.7m
7. Cumulative profit = $1.8 + 0.7 = \text{USD } 2.5\text{m}$, not the USD 3.0m originally expected

Y1: rev 6.0m, profit 1.8m · Y2: rev 4.0m, profit 0.7m · cumulative 2.5m

¹ Revenue = cost of sales recognised as incurred, plus a proportionate share of the estimated profit. Progress measured by an input method (cost-to-cost).

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 1 "Long-Term Contracts" (Armored Vehicles Inc.), pp. 43–44.

Principal versus agent cuts reported revenue by 70% and lifts net margin from 20% to 67% — with identical cash profit

A principal controls the good before it transfers and reports GROSS revenue. An agent merely arranges the transfer and reports only its NET fee or commission.

Line item	MegaDigital as PRINCIPAL			MegaDigital as AGENT		
Sales	USD 100	•	100%	USD 30	•	100%
Cost of sales	USD 70	•	70%	USD 0	•	0%
Gross profit	USD 30	•	30%	USD 30	•	100%
SG&A	USD 10	•	10%	USD 10	•	33%
Net profit	USD 20	•	20%	USD 20	•	67%

THE TEST — WHO CONTROLS THE GOOD?

Principal if the entity controls the good or service before it transfers to the customer: it sets the price, holds the inventory risk, and is primarily responsible for fulfilment. Otherwise it is an agent, and only the commission is revenue.

WHY ANALYSTS CARE

Marketplaces, ride-hailing, travel and ad-tech all run a mix of principal and agent sales. Growth rates and margins are not comparable across peers — or across time in the same company — unless the mix is disclosed and tracked.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 1 "Principal Versus Agent" (MegaDigital), pp. 41–42.

Expense recognition follows one of three models: match to the related revenue, expense as incurred, or capitalize and amortize over useful life

The choice turns on when the economic benefit is consumed: alongside the revenue it produced, in the current period regardless, or across several future periods.

1 · MATCHING

Recognise the expense in the same period as the revenue it produced.

Cost of goods sold is the archetype — the Kahn example on the next slide is exactly this.

IFRS avoids the phrase "matching principle" and speaks of a matching concept. The mechanic is identical.

2 · PERIOD — AS INCURRED

Expense when consumed, with no attempt to link it to revenue.

Administrative and managerial salaries, IT, rent, utilities, advertising, repairs and maintenance, research.

Used when the benefit is used up in the period and no revenue link is identifiable.

3 · CAPITALIZE + AMORTIZE

Park the cost on the balance sheet, then release it over the asset's useful life.

PP&E to depreciation, intangibles to amortization, development-phase software.

Also an investing cash outflow, not an operating one — which is where the mischief starts.

Matching, not payment, sets cost of goods sold — and which units you call "sold" moves gross profit by a quarter

Kahn buys 7,600 units in four rising-price lots and sells 5,600 at USD 50. Identical cash out and cash in — three different reported gross profits.

Kahn Distribution Ltd — 20X1	Specific ID ¹	Weighted-avg	LIFO ²
Purchases — 7,600 units	USD 321,600	USD 321,600	USD 321,600
Revenue — 5,600 × USD 50	USD 280,000	USD 280,000	USD 280,000
Cost of goods sold	USD 231,800	USD 236,968	USD 241,600
Gross profit	USD 48,200	USD 43,032	USD 38,400
Ending inventory (2,000 units)	USD 89,800	USD 84,632	USD 80,000
Gross margin	17.2%	15.4%	13.7%

THE LOTS, AND WHY THE COLUMNS DIFFER

Q1 2,000 @ 40 · Q2 1,500 @ 41 · Q3 2,200 @ 43 · Q4 1,900 @ 45 = 7,600 units for USD 321,600. Weighted-average unit cost = $321,600 / 7,600 = \text{USD } 42.32$. LIFO leaves the cheap Q1 units on the balance sheet and expenses the dear ones — USD 9,800 less gross profit on identical cash flows.

¹ Curriculum solution uses specific identification: 1,900 unsold units from Q4, 100 from Q3. With these lots that gives the same answer as FIFO.

² LIFO is permitted under US GAAP only; IFRS allows FIFO or weighted average. Cost-flow methods are developed in the Inventories module.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 2 "The Matching of Inventory Costs with Revenues" (Kahn Distribution Ltd), pp. 45–46.

Capitalizing an expenditure instead of expensing it lifts first-year profit and ROE — and guarantees you give every euro of it back later

Two identical companies: EUR 1,000 of equity, revenue 1,500 a year, cash costs 500, one EUR 900 machine with a three-year life, 30% tax. CAP capitalizes it; NOW expenses it.

EUR	CAP Y1	CAP Y2	CAP Y3	NOW Y1	NOW Y2	NOW Y3
Revenue	1,500	1,500	1,500	1,500	1,500	1,500
Cash expenses	(500)	(500)	(500)	(1,400)	(500)	(500)
Depreciation	(300)	(300)	(300)	—	—	—
Net income	490	490	490	70	700	700
Cash from operations	790	790	790	70	700	700
Net profit margin	33%	33%	33%	5%	47%	47%
ROE	39%	28%	22%	7%	49%	33%

WHAT THIS IS REALLY ABOUT

Three-year totals are identical: EUR 1,470 of net income and EUR 1,470 of cash either way. Only the shape differs — and CAP books the EUR 900 as investing, so its cash from operations looks EUR 900 better.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 3 and Exhibit 1 "Capitalizing versus Expensing" (CAP Inc. vs NOW Inc.), pp. 47–49.

SECTION 03

Non-recurring items & EPS

Strip out what will not repeat – discontinued operations, unusual items, policy changes – before you forecast anything

Six distinct treatments, one question behind all of them: is this item going to happen again? The curriculum warns against the lazy answer of ignoring every unusual item.

Item	Accounting treatment	Analyst action
Discontinued operations	Net of tax, own line below continuing ops	Exclude when forecasting
Unusual or infrequent items	Inside continuing ops but shown separately	Case by case; often strip
Change in accounting POLICY	Retrospective – prior periods restated	Use the restated series
Change in accounting ESTIMATE	Prospective only – no restatement	Repeat changes = red flag
Prior-period error	Restate all prior periods presented	Check the control note
Change in scope / FX	No disclosure required – issuers volunteer	Ask for organic growth

FROM REPORTED TO CORE EARNINGS

Reported net income, less after-tax discontinued operations, less one-off gains, plus one-off charges = core continuing earnings. That is the number multiples are applied to. Danone's "exceptional" note is the curriculum's illustration: disposal gains, arbitration receipts, integration costs and intangible impairments, all in one line.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, "Non-Recurring Items", pp. 63–68.

Basic EPS is straightforward division — the only tricky piece is the weighted-average number of shares outstanding over the period

Each change in the share count is weighted by the fraction of the year for which that count was actually outstanding. Splits and stock dividends are the exception — they are retroactive.

$$\text{Basic EPS} = (\text{Net income} - \text{Preferred dividends}) / \text{WA shares}$$

where: WA = weighted average common shares outstanding. Preferred dividends come out whether or not they were declared as an obligation of the period.

ANGLER PRODUCTS • YEAR ENDED 31 DECEMBER 2018

Net income USD 2,500,000. Preferred dividends declared and paid USD 200,000. Share count moves twice during the year.

- 1 Jan: 1,000,000 shares outstanding $\times 3/12 = 250,000$
2. 1 Apr issue of +200,000 $\rightarrow 1,200,000 \times 6/12 = 600,000$
3. 1 Oct buyback of -100,000 $\rightarrow 1,100,000 \times 3/12 = 275,000$
4. Weighted average = 250,000 + 600,000 + 275,000 = 1,125,000 shares
5. Basic EPS = $(2,500,000 - 200,000) / 1,125,000 = \text{USD } 2.04$

Basic EPS = USD 2.04 • note the closing count is 1,100,000, not 1,125,000

¹ Example 10 extends this: a 2-for-1 split on 1 December is treated as if it happened on 1 January $\rightarrow 2,250,000$ weighted shares and basic EPS of USD 1.02.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 9 "A Basic EPS Calculation (2)" and Exhibit 13 (Angler Products), p. 71.

The if-converted method adds back the interest saved net of tax AND the new shares, then recomputes — both ends of the fraction move

Always run the antidilution check. If the "diluted" figure comes out HIGHER than basic, the security is antidilutive, it is excluded, and diluted EPS equals basic EPS.

$$\text{Diluted EPS} = [\text{NI} + \text{interest} \times (1 - t)] / [\text{WA} + \text{new shares}]$$

where: Numerator rises by the after-tax interest the company would not have paid; denominator rises by the shares it would have issued.

OPPNOX COMPANY • CONVERTIBLE BOND • YEAR ENDED 31 DECEMBER 2018

Net income USD 750,000. Weighted-average shares 690,000. One potentially dilutive security: USD 50,000 of 6% convertible bonds, convertible into 10,000 shares. Tax rate 30%.

1. Interest saved if converted: $50,000 \times 6\% = \text{USD } 3,000$
2. After-tax add-back: $3,000 \times (1 - 0.30) = \text{USD } 2,100 \rightarrow$ numerator USD 752,100
3. Denominator: $690,000 + 10,000$ conversion shares = 700,000
4. Basic EPS = $750,000 / 690,000 = \text{USD } 1.09$
5. Diluted EPS = $752,100 / 700,000 = \text{USD } 1.07 \rightarrow$ below basic, so dilutive: report it

Basic USD 1.09 • Diluted USD 1.07 • both ends of the fraction moved

¹ Convertible preferred: add back the preferred dividend, not interest. Example 11 — Bright-Warm Utility, USD 3.10 basic $\rightarrow$ USD 2.92 diluted.

² Example 15 (Dim-Cool Utility): if-converted gives USD 3.13 against basic USD 3.10 — antidilutive, excluded, reported diluted EPS stays USD 3.10.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 12 and Exhibit 15 (Oppnox Company), p. 74.

Options dilute through the treasury-stock method: shares issued at the strike, and the cash raised is assumed spent buying stock back at market

The numerator never moves — exercising an option does not change net income. Only the share count does, and only by the net amount.

$$\text{Incremental shares} = N \times (1 - \text{Strike} / \text{Avg market})$$

where: N = options outstanding. $30,000 \times (1 - 35/55) = 10,909$ — the same answer as the long route through proceeds and buy-back.

HIHOTEC COMPANY • YEAR ENDED 30 JUNE 2018 • 30,000 OPTIONS

Net income USD 2,300,000. Weighted-average shares 800,000. Options outstanding from the start of the year: 30,000 at a USD 35 exercise price. Average market price over the year USD 55.

1. Assume exercise: 30,000 new shares issued at the USD 35 strike
2. Cash proceeds = $30,000 \times 35 = \text{USD } 1,050,000$
3. Shares repurchased at the USD 55 average market price = $1,050,000 / 55 = 19,091$
4. Incremental shares = $30,000 - 19,091 = 10,909$; numerator unchanged
5. Basic = $2,300,000 / 800,000 = \text{USD } 2.88$; diluted = $2,300,000 / 810,909 = \text{USD } 2.84$

Basic USD 2.88 • Diluted USD 2.84 • denominator only, numerator untouched

¹ IFRS gets there differently — proceeds are treated as issuing 19,091 "inferred" shares at market; $30,000 - 19,091 = 10,909$. Same answer (Example 14).

² Out-of-the-money options are antidilutive and ignored: if the strike exceeds average market price the formula returns a negative number.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Example 13 and Exhibit 16 (Hihotech Company), pp. 75–76.

SECTION 04

Balance sheet: classification & measurement

The current / non-current split is what makes liquidity and solvency ratios computable – misclassify one line and all of them move

Current if it will be realised or settled within twelve months or within one normal operating cycle, whichever is longer. Everything else is non-current.

Category	Current	Non-current
Assets	Cash · marketable securities · A/R · inventory	PP&E, intangibles, goodwill
Liabilities	A/P · short-term debt · current LT debt · accruals	LT debt, leases, pensions
Equity	Not split – sits below all liabilities	Share capital, RE, AOCI

LIQUIDITY – EXHIBIT 13

Current = Current assets / Current liabilities

Quick = (Cash + Mkt sec + Receivables) / CL

Cash = (Cash + Mkt sec) / CL

SAP 2017: 1.17 · 1.07 · 0.49 — all three below the 2016 levels of 1.20 · 1.11 · 0.50.

SOLVENCY – EXHIBIT 13

Debt-to-equity = Total debt / Total equity

LT debt-to-equity = LT debt / Total equity

Total debt = Total debt / Total assets

Financial leverage = Total assets / Total equity

SAP 2017: D/E 25.8%, leverage 1.66× — both improved on 2016.

Source: CFA Institute, Curriculum Vol 4, Learning Module 3, Exhibit 13 "Balance Sheet Ratios", p. 114; SAP and Apple balance sheets, Exhibits 2, 3, 6 and 7, pp. 96–103.

Not every balance-sheet line is measured the same way — historical cost, amortized cost and fair value coexist on the same page

A balance sheet total is not a valuation. It is three different questions answered in three different ways and then added together — know which one you are reading.

HISTORICAL COST

What was paid on acquisition, never revisited for value.

Land, PP&E at gross, inventory at cost.

Answers "what did we pay?"

Verifiable and auditable. Can be decades stale — a plant bought in 1985 sits at 1985 money.

AMORTIZED COST

Initial amount, less principal repaid, plus or minus amortisation of discount or premium, less impairment.

Held-to-maturity debt, loans, notes receivable, most debt issued.

Answers "what is still owed on the original terms?"

Drives the effective interest rate.

FAIR VALUE

Price received to sell an asset, or paid to transfer a liability, in an orderly transaction at the measurement date.

Trading securities, equity investments, derivatives.

Answers "what is it worth today?"

Level 1 / 2 / 3 inputs must be disclosed.

Financial assets sort into three measurement buckets – amortized cost, FVOCI, FVPL – and only one of them lets market moves reach net income

Two tests decide the bucket: what is the business model for holding the asset, and are the contractual cash flows solely principal and interest?

Bucket	IFRS 9 test	What hits P&L
Amortized cost	Hold to collect + cash flows solely P&I	Interest only
Fair value through OCI	Hold to collect and sell; equity by election	Interest; FV → OCI
Fair value through P&L	Everything else – trading, non-SPPI, elected	FV move hits income

US GAAP EQUIVALENTS

Held-to-maturity ≈ amortized cost.
Available-for-sale, DEBT only ≈ FVOCI.
Trading debt ≈ FVPL.

Under US GAAP every equity investment goes to FVPL — there is no FVOCI election for equity. That single difference is what moves reported earnings volatility.

EXHIBIT 5 – ONE BOND, THREE ANSWERS

EUR 100m bond at 5%. Six months in: EUR 2.5m coupon received, market value up EUR 2.0m.

Amortized cost → P&L 2.5m, asset 100m
FVOCI → P&L 2.5m, OCI 2.0m, asset 102m
FVPL → P&L 4.5m, asset 102m

Source: CFA Institute, Curriculum Vol 4, Learning Module 3, Exhibits 4 and 5, "Financial Instruments", pp. 101–102. IFRS 9 / IAS 32.

SECTION 05

Intangibles, goodwill & common-size

Internally developed intangibles are expensed under US GAAP but capitalized under IFRS once the development-phase criteria are met

Two technology companies with identical R&D spend but different domiciles will report different operating margins, different total assets and different ROA — forever.

IFRS — IAS 38

Research phase: expense.

Development phase: capitalise once all six criteria are met.

Finite life → amortise. Indefinite life → no amortisation, annual impairment test.

Cost OR revaluation model.

US GAAP — ASC 730 / 350

All internally developed intangibles and R&D expensed as incurred.

Exception: software development costs, once technological feasibility is established.

Cost model only — no revaluation.

EXPENSED UNDER BOTH

Internally generated brands, mastheads, publishing titles, customer lists · start-up costs · training · administrative and general overhead · advertising and promotion · relocation and reorganisation · redundancy costs.

EXHIBIT 1 — ALPHA INC., TWO PROJECTS (EUR 000s)

Project 1 — a steering mechanism that reacts to finger impulses: still seeking new knowledge, so RESEARCH. Nothing capitalised under either framework.

Project 2 — a prototype electronic welding rig, already judged technically feasible, saleable and producible: DEVELOPMENT. Under IFRS capitalise materials 620 + direct labour 320 + design, construction and testing 470 = EUR 1,410. Administrative personnel costs are never capitalised. Under US GAAP: zero, all expensed.

¹ IAS 38 requires all six: technical feasibility, intent to complete, ability to use or sell, probable benefits, resources, reliable measurement.

² Purchased intangibles are capitalised under both frameworks — the divergence concerns only internally generated ones.

Source: CFA Institute, Curriculum Vol 4, Learning Module 3, "Intangible Assets" and Exhibit 1 "Summary of Expenses" (Alpha Inc.), pp. 93–95. IAS 38.

Goodwill is the residual when a buyer pays more than the fair value of identifiable net assets — impairment is where it comes back

Accounting goodwill only ever arises in an acquisition. Economic goodwill — the present value of excess returns — never touches the balance sheet; it lives in the share price.

$$\text{Goodwill} = \text{Cost to purchase} - \text{Net identifiable assets}$$

where: Net identifiable assets = fair value of identifiable tangible and intangible assets, less fair value of liabilities and contingent liabilities assumed.

THE THREE STEPS — ILLUSTRATIVE

Step 1 — cost to purchase the target: USD 800m.

Step 2 — fair value of identifiable assets 600m, less liabilities and contingent liabilities assumed 150m = net identifiable assets 450m.

Step 3 — goodwill = $800 - 450 = \text{USD } 350\text{m}$, 44% of the price.

Pay less than 450 and it is a bargain purchase: the gain goes straight to P&L.

SAFEGWAY, Q4 2009 — WHEN IT UNWINDS

Non-cash goodwill impairment of USD 1,974.2m, USD 1,818.2m net of tax. Reported net loss USD 1,609.1m, or USD 4.06 per diluted share. Without the charge: net income of USD 209.1m, USD 0.53 per share.

That is 82.6% of opening goodwill and 11.3% of opening total assets — and it says the acquisitions were worth less than what was paid for them.

¹ Goodwill is never amortised. It is tested for impairment at least annually, at the cash-generating-unit (IFRS) or reporting-unit (US GAAP) level.

² Many analysts strip goodwill out of the balance sheet for ratios and strip impairment out of earnings when reading operating trends.

Source: CFA Institute, Curriculum Vol 4, Learning Module 3, "Goodwill" — three-step recognition and the Safeway Q4 2009 impairment, pp. 97–99. IFRS 3 / ASC 805.

Common-size statements strip out scale, and what is left is strategy — C is a fifth of A's size and structurally identical

Three companies in one industry. A and B each do USD 10m of sales; C does USD 2m. Every line as a percentage of sales — and the absolute numbers stop mattering.

USD	A	A %	B	B %	C	C %
Sales	10,000,000	100%	10,000,000	100%	2,000,000	100%
Cost of sales	3,000,000	30%	7,500,000	75%	600,000	30%
Gross profit	7,000,000	70%	2,500,000	25%	1,400,000	70%
SG&A	1,000,000	10%	1,000,000	10%	200,000	10%
Research & development	2,000,000	20%	—	0%	400,000	20%
Advertising	2,000,000	20%	—	0%	400,000	20%
Operating profit	2,000,000	20%	1,500,000	15%	400,000	20%

WHAT THE PERCENTAGES SAY

A and C are the same business at different scale — identical on every line, so C's smaller profit is not worse performance. A versus B is strategy: 20% R&D plus 20% advertising buys A a 70% gross margin; B spends nothing and gets 25%.

Source: CFA Institute, Curriculum Vol 4, Learning Module 2, Exhibit 18 "Income Statement for Three Hypothetical Companies", pp. 79–80.

Scale every line to total assets and strategy becomes visible — Apple grew organically, Microsoft bought growth

Two companies in the same industry, both asset-light, both cash-rich, both similarly levered. One row — goodwill — separates their entire approach to growth.

% of total assets	Apple	Microsoft	What it says
Cash & short-term securities	19.8	55.2	Both hoard — Microsoft short-dated
Long-term marketable securities	51.9	2.5	Apple parks a decade of cash flow
Accounts receivable	4.8	8.2	Apple sells direct, and for cash
Total current assets	34.3	66.3	Very different, both very liquid
Property, plant & equipment	9.0	9.8	Both asset-light, at sector median
Goodwill	1.5	14.6	Organic growth versus acquisition
Acquired intangibles	0.6	4.2	Same story, second line
Deferred revenue (current)	2.0	14.1	Microsoft is a subscription book
Total liabilities	64.3	70.0	Leverage is not the difference
Total shareholders' equity	35.7	30.0	Total assets = 100.0 for both

Source: CFA Institute, Curriculum Vol 4, Learning Module 3, Example 1 and Exhibit 12 "Cross-Sectional Analysis" (Apple FY2017, Microsoft FY2017), pp. 112–113.

Deck 6 — FSA I — Framework, Income Statement & Balance Sheet

- FSA is a SIX-phase discipline — purpose, collect, process, analyze, conclude and communicate, follow up — drawing on four regulated layers: filings, notes, MD&A, auditor report.

- IFRS 15 is one five-step model: contract → obligations → price → allocate → recognize. Armored Vehicles at 60% complete books 6.0m of revenue and 1.8m of profit — until the cost estimate moves.

- Expense recognition is match, expense as incurred, or capitalize and amortize. CAP and NOW earn the same EUR 1,470 over three years — capitalizing only changes the shape and flatters operating cash flow.

- The balance sheet is a mixed-measurement model: historical cost, amortized cost, fair value. IFRS 9 sorts financial assets by business model plus the SPPI test into amortized cost, FVOCI and FVPL.

- IFRS and US GAAP are converged on revenue and leases but still diverge on LIFO, inventory write-down reversal, interest paid in the cash flow statement, development costs and revaluation.

- Principal versus agent cuts MegaDigital's revenue from 100 to 30 and lifts net margin from 20% to 67%, with identical cash profit. Know the mix before comparing any marketplace.

- Basic EPS weights share counts by time (Angler: 1,125,000, EPS 2.04). Diluted adds back after-tax interest and conversion shares (Oppnox: 1.09 → 1.07) or net option shares (Hihotech: 2.88 → 2.84) — only when dilutive.

- IFRS capitalizes development-phase costs once six criteria are met (Alpha: EUR 1,410); US GAAP expenses all of it. Goodwill = cost less net identifiable assets, never amortized, tested annually — Safeway wrote off 82.6% of it in one quarter.